

Classwork 2

Exponentials and logarithms

Grades: [A] [M] [E]

Show all working. Give answers to 3 decimal places unless the question asks for an exact answer.

Part A · Switch, combine, solve

A1 Evaluate $\log_3 243$. [A]

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A2 Solve $\log_6 x = 2$. [A]

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A3 Write $3 \log 2 + \log 5 - \log 4$ as a single logarithm, and find its value. [A]

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A4 Solve $6^x = 75$. [A]

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A5 Solve $\log_x 16 = 4$. [A]

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Part B · Two logs, and a model

B1 Solve $2^{3x-2} = 50$. [M]

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B2 Solve $\log_5(x+1) + \log_5(x-3) = 1$. [M]

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B3 A phone bought for \$1200 loses 18% of its value each year. When is it first worth less than \$300? [M]

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Part C · A quadratic in disguise

C Solve $4^x - 5 \times 2^x + 4 = 0$. Explain each step.

[E]

Words you may use

let $u = 2^x \cdot 4^x = (2^x)^2$ · factorise · change back · always positive

Sentence starters (optional): *Since $4 = 2^2$, ...* *Letting $u = 2^x$, ...* *So $2^x = \dots$*

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Extension

Solve $2^x = 3^{x-1}$.

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